GENOMIC EPIDEMIOLOGY: STUDY DESIGNS

17 / 02 / 26

Sandra Perdomo
IARC

Data
Discovery
Prevention
Knowledge

60 years | International Agency for Research on Cancer

Cancer research into action

EPIDEMIOLOGY

Study of the incidence, distribution and determinants (causes and effects) of diseases in human populations.

Objectives:

- Identify pathologies with a high incidence in certain regions
- Identify associated risk factors
- To provide knowledge for formulating public policies that can create effective prevention measures

EPIDEMIOLOGY

Incidence/Mortality

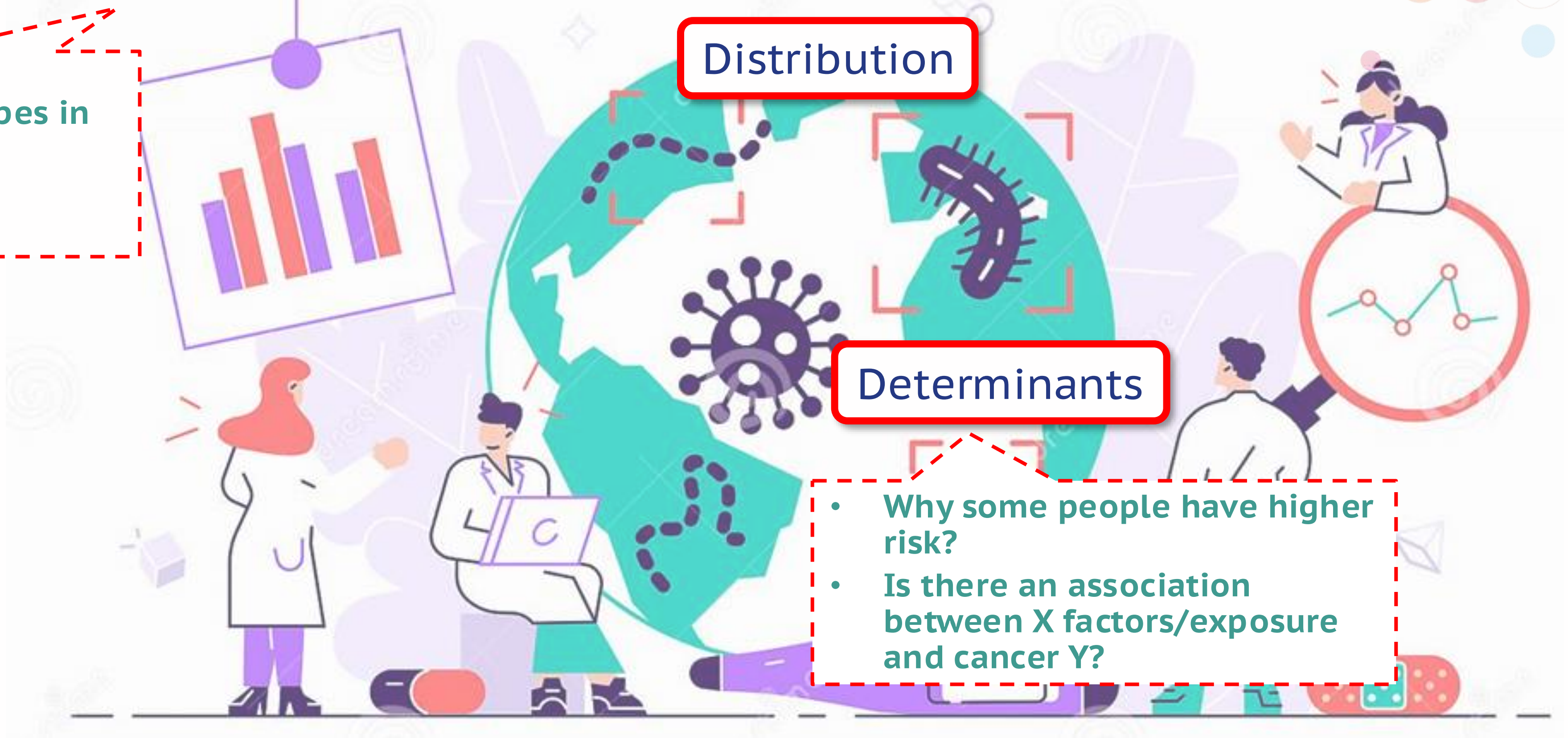
- What are the most frequent cancer types in this population?
- How many new cases/year?

Distribution

- Who has more risk? (populations, age groups, etc)
- Has the frequency of cancer X changed over the xx years?
- How does cancer vary from one place to another?

Determinants

- Why some people have higher risk?
- Is there an association between X factors/exposure and cancer Y?



Martel et al., Lancet; 2020

BEST STUDY DESIGNS (?)

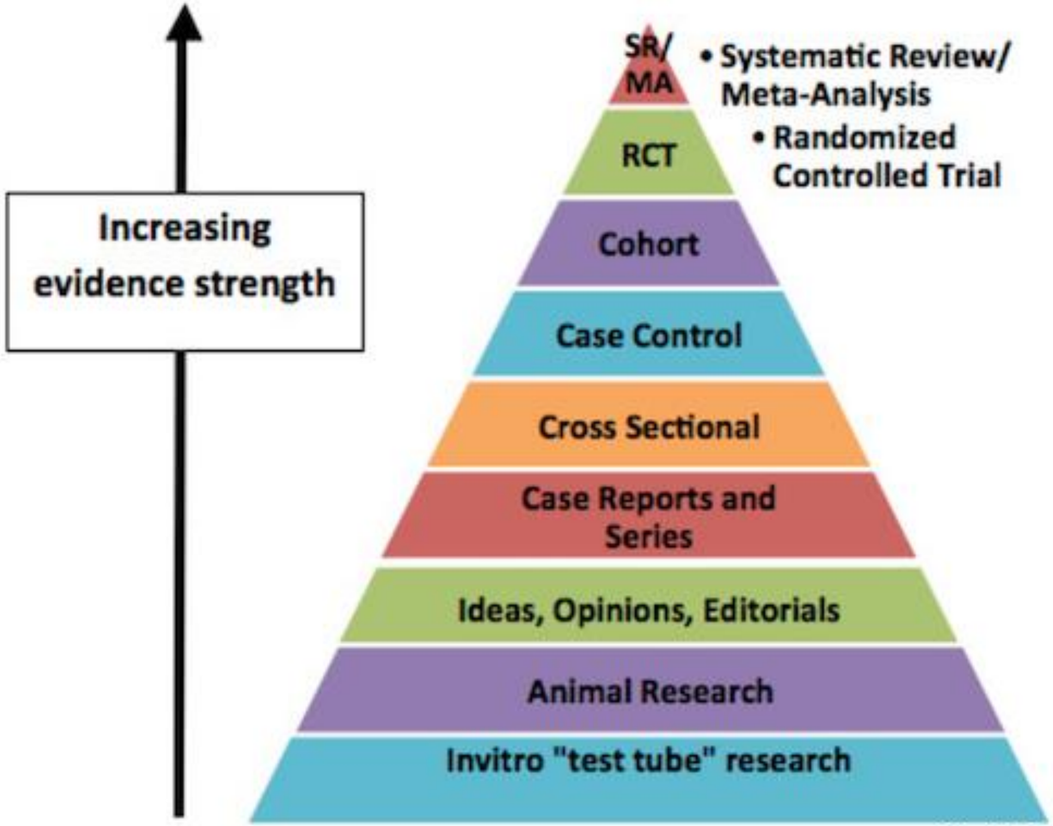
-No perfect design

-Important to have a clear idea of the research question

-Awareness of required data/samples and limitations

-Planning (pilots)

-Resources



Credit: Univ of New Hampshire/
SUNY Downstate

Adaptation of the Evidence Pyramid Diagram developed by the Medical Research Library of Brooklyn, SUNY Downstate Medical Center.

SUMMARY OF STUDY DESIGNS

-Cross-sectional and short- term longitudinal studies

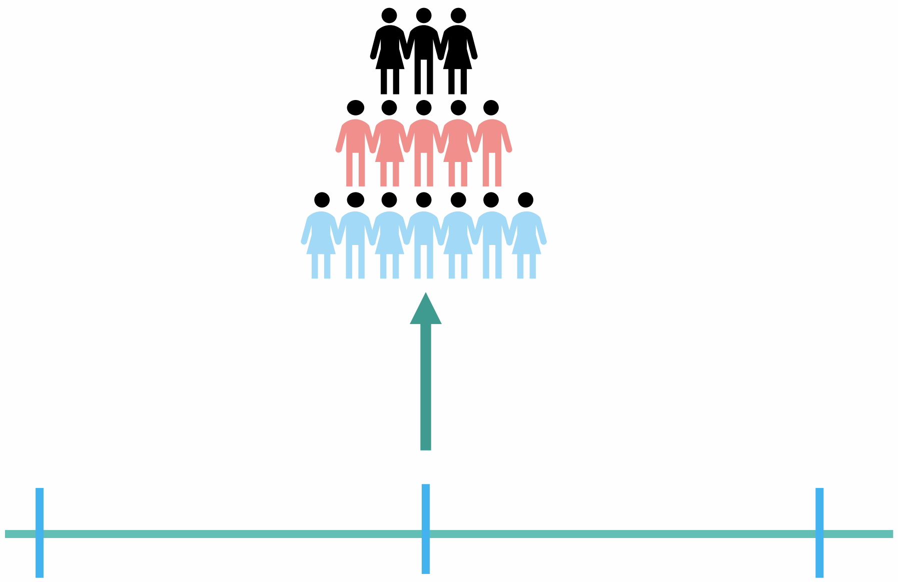
-Prospective cohort studies: nested case-control, case-cohort

-Case-control studies

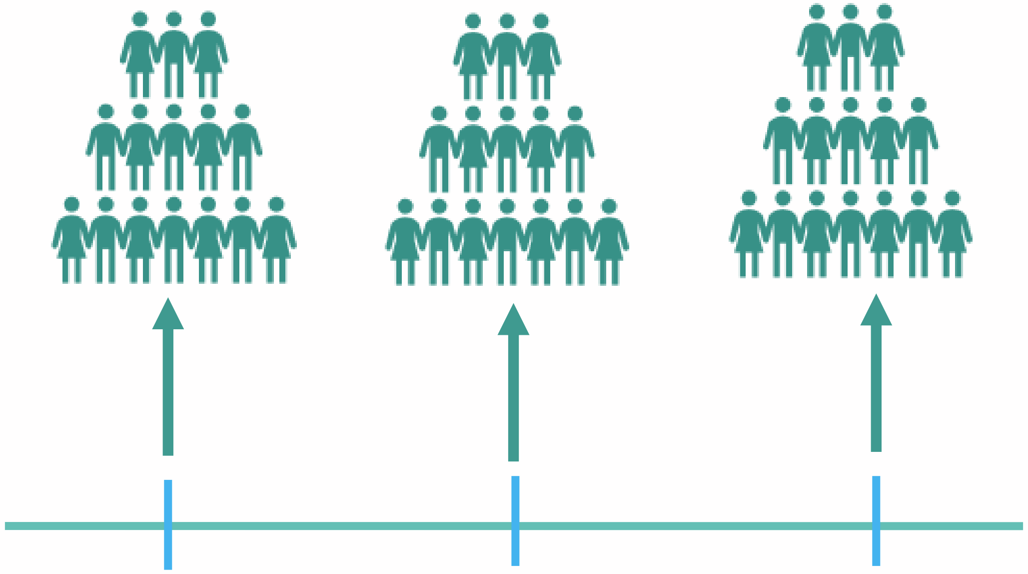
-Case only studies (case series)

CROSS-SECTIONAL

SHORT- TERM LONGITUDINAL STUDIES



TIME



TIME

- Information at one timepoint (Snapshot):
- -Exposures variables
- -Intermediate endpoints

Information at various timepoints in a short period:

-Exposures variables (healthy individuals)

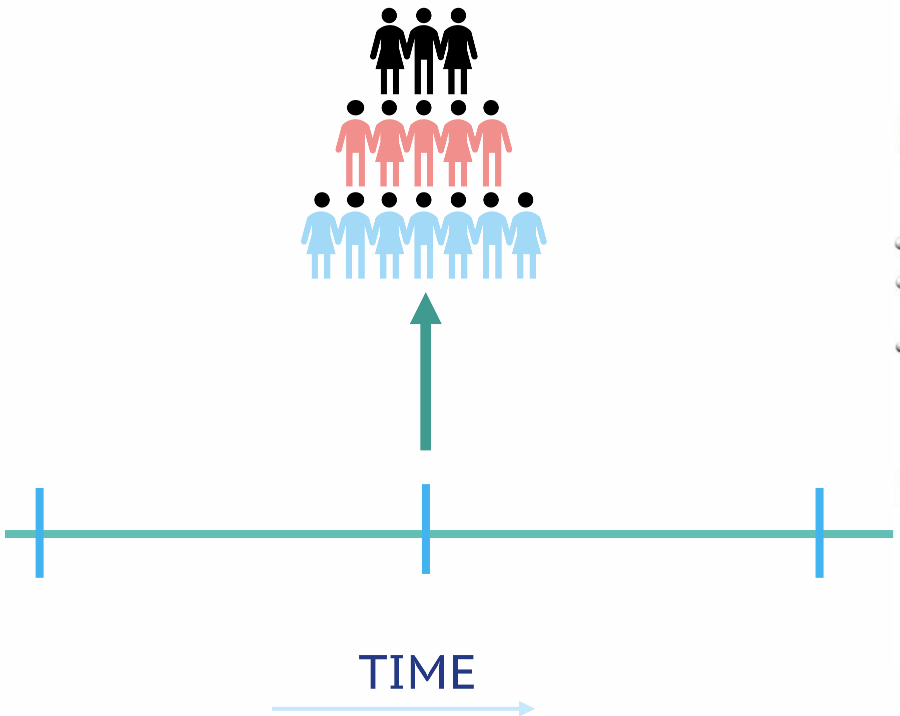
-Intermediate endpoints

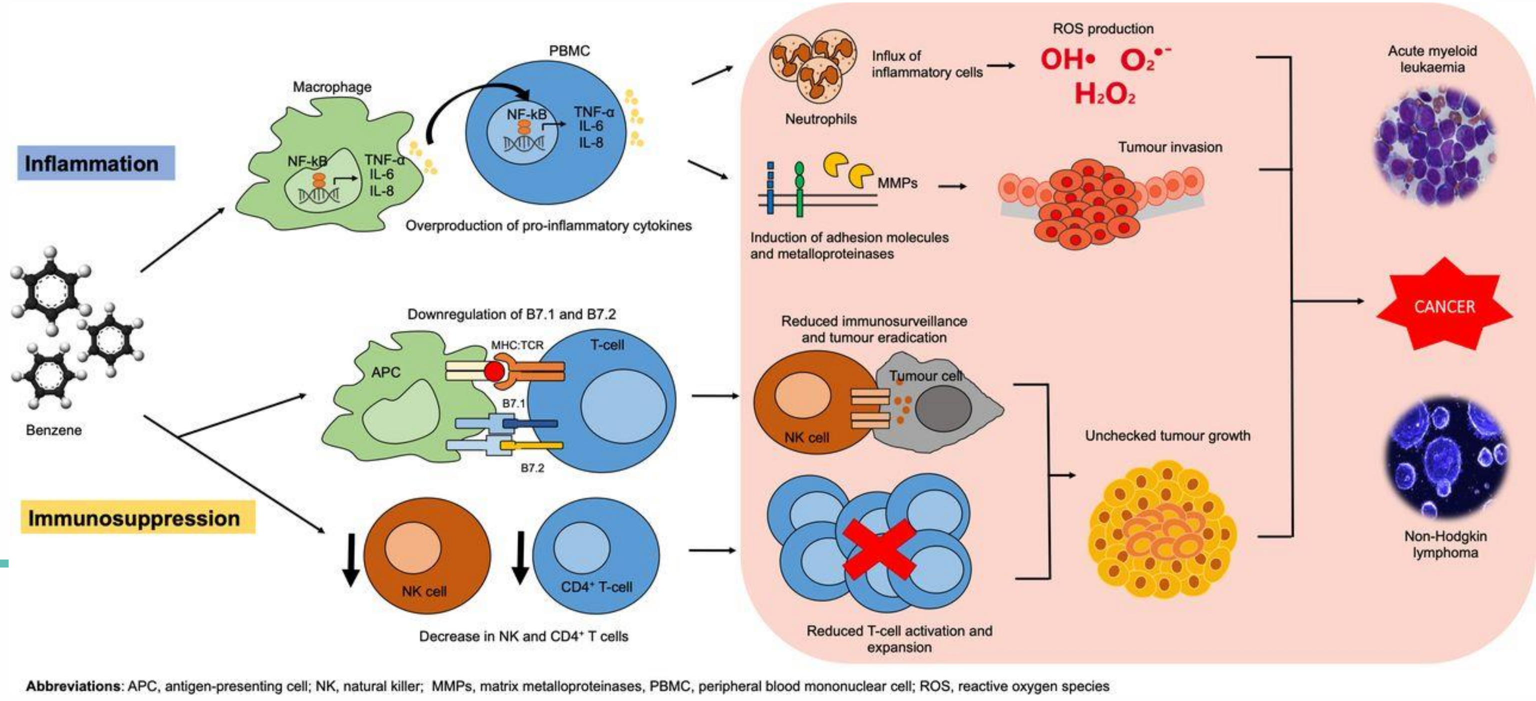
-Disease at a specific time point

60 years International Agency for Research on Cancer

Cancer research into action

CROSS-SECTIONAL STUDY OF BENZENE AND LEUKEMIA



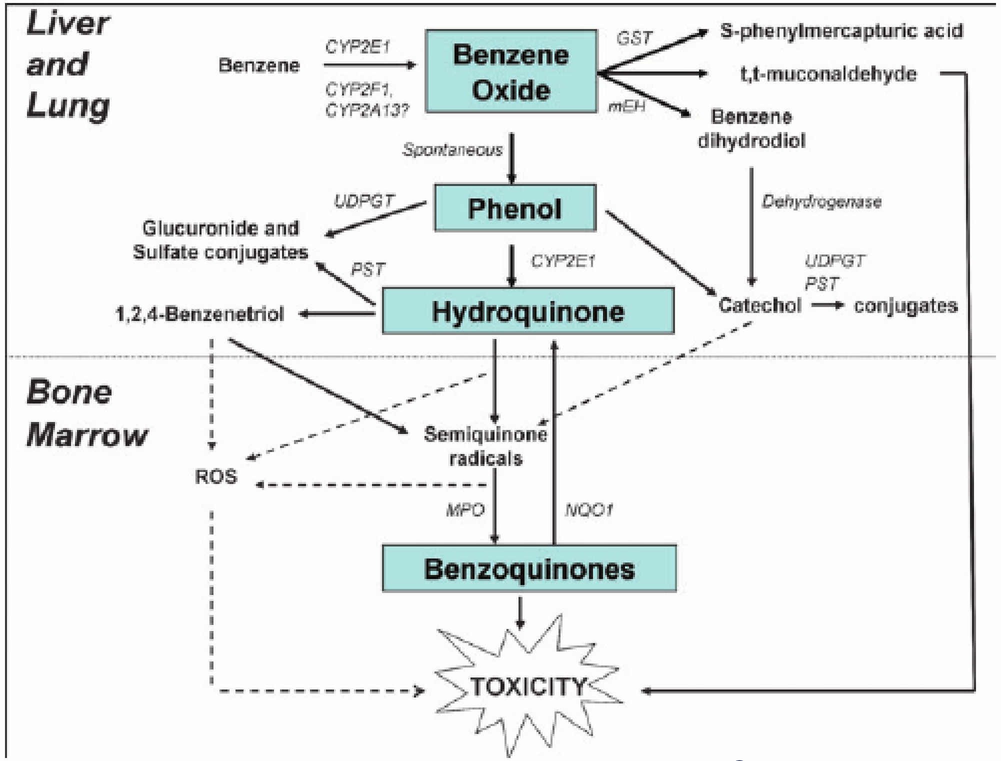


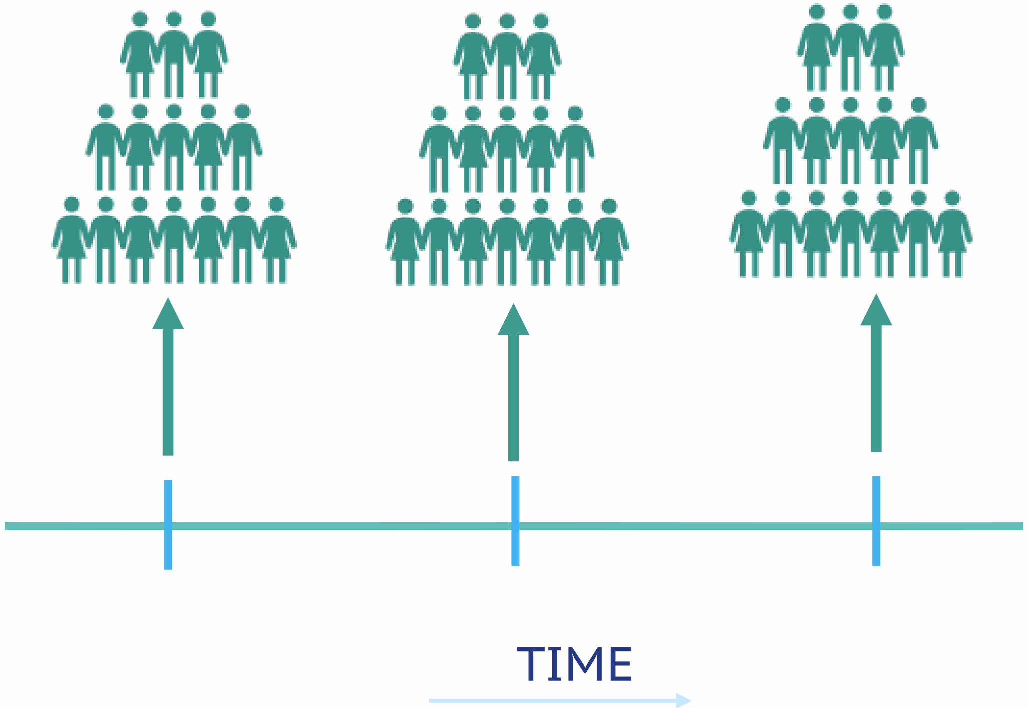
- Information at one timepoint (Snapshot):
- -Occupational exposure to Benzene
- -Hematotoxicity-measure of levels of benzene and cell counts

Guo H, et al. Occupational and Environmental Medicine 2021;78:377-384.

EFFECT OF BENZENE LEVEL EXPOSURE AND LEUKEMIA

Benzene=carcinogenic to humans (Group 1)





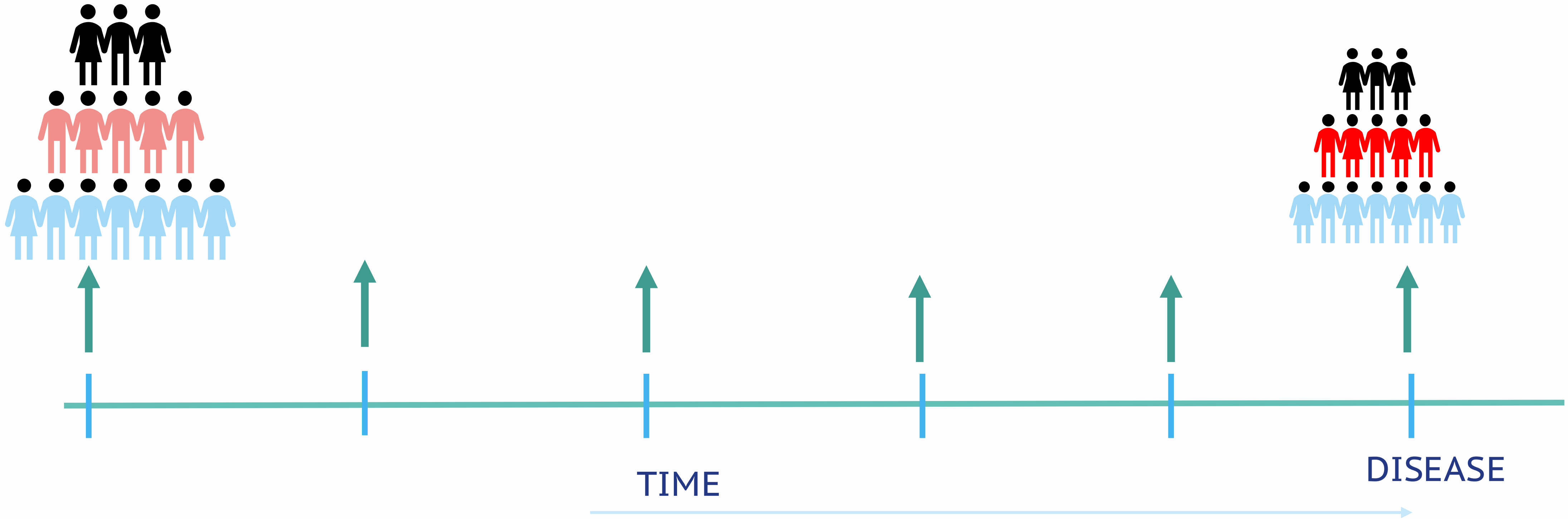
Information at various timepoints in a short period:

-Dynamics of the low Benzene exposure 1 ppm

-Level of toxicity in deferent timepoints, increase level of DNA aberrations

International Agency for Research on Cancer
Cancer research into action

PROSPECTIVE COHORT STUDIES: NESTED CASE-CONTROL, CASE-COHORT

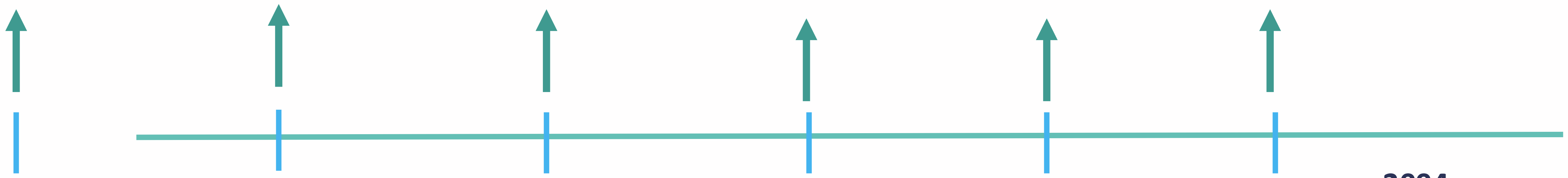


- Information on a group of individuals over time
- Exposure variables, biological samples at multiple time points
- Outcome of interest: risk of disease (multiple disease outcomes)
- Size, sampling designs

PROSPECTIVE COHORT STUDIES

Nurses' Health Study

Questionnaire
Blood
Serum
Urine
DNA from oral swabs



1976 121,700 women 30-50 yrs

to investigate the potential long-term consequences of oral contraceptives

TIME

DISEASE **2004**

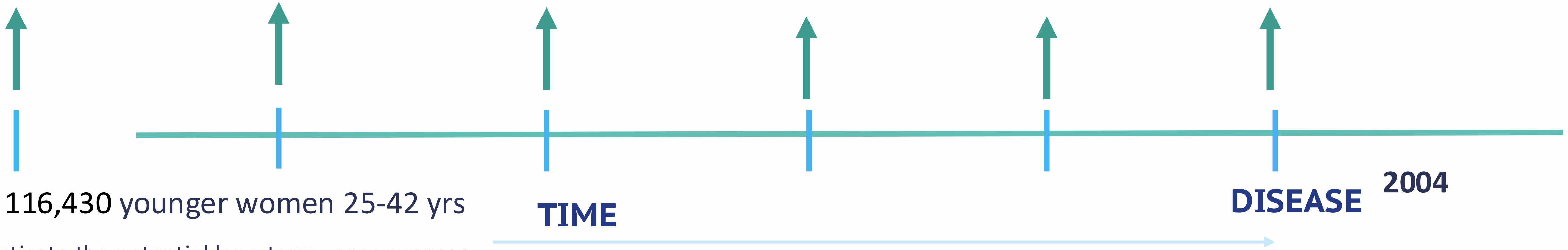
- Cigarette smoking and cardiovascular disease
- postmenopausal obesity and breast cancer.
- The impact of diet on health in the Nurses' Health Studies has informed dietary recommendations, including national dietary guidelines.

https://nurseshealthstudy.org

PROSPECTIVE COHORT STUDIES

Nurses' Health Study

Nurses' Health Study II (NHS II)



1989 116,430 younger women 25-42 yrs

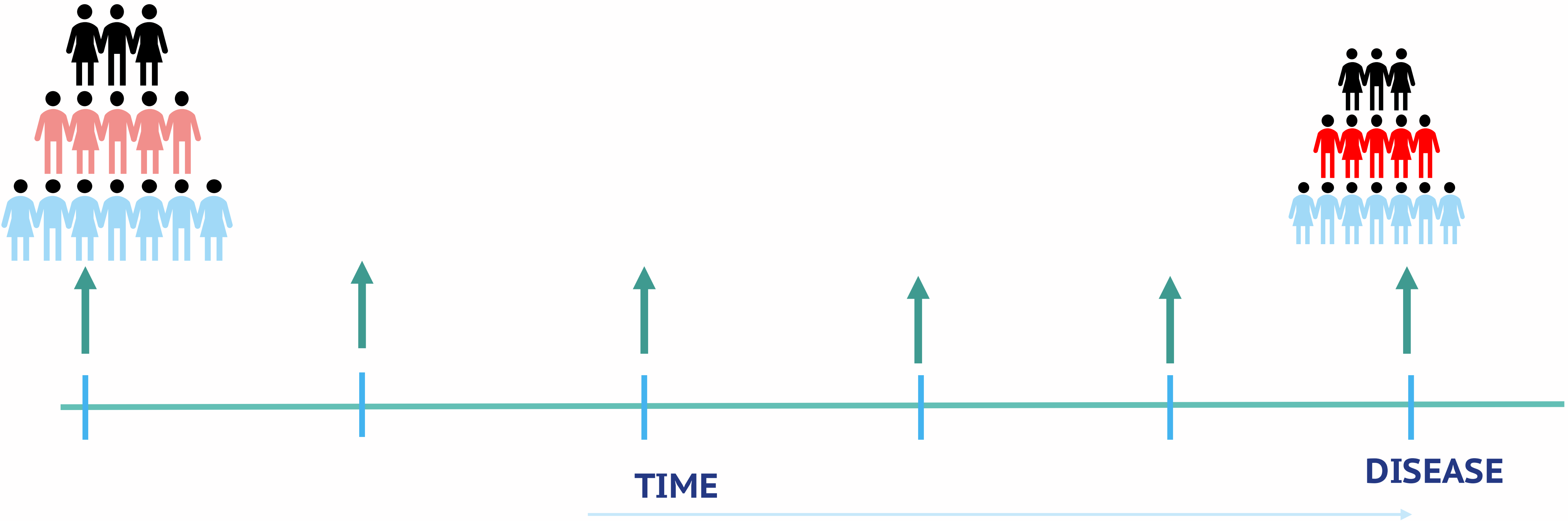
to investigate the potential long-term consequences of oral contraceptives

- Cigarette smoking and cardiovascular disease
- postmenopausal obesity and breast cancer.
- The impact of diet on health in the Nurses' Health Studies has informed dietary recommendations, including national dietary guidelines.

PROSPECTIVE COHORT STUDIES

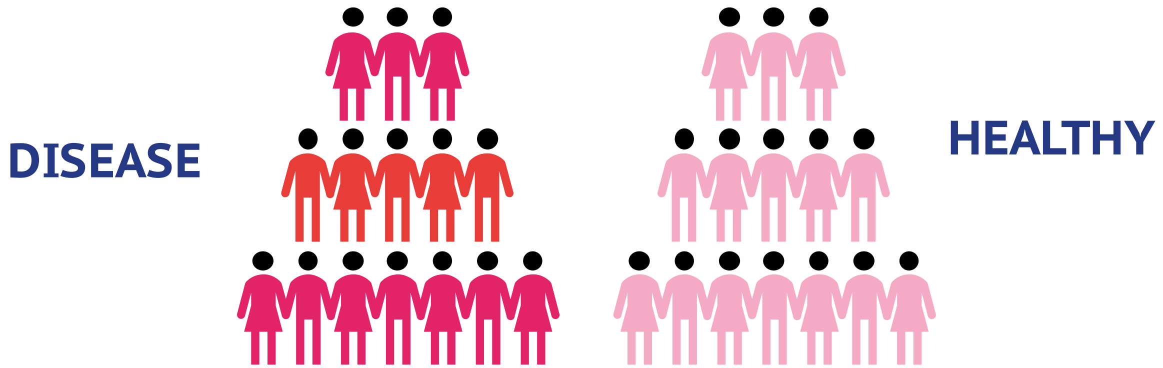
Nested case-control

Case Cohort



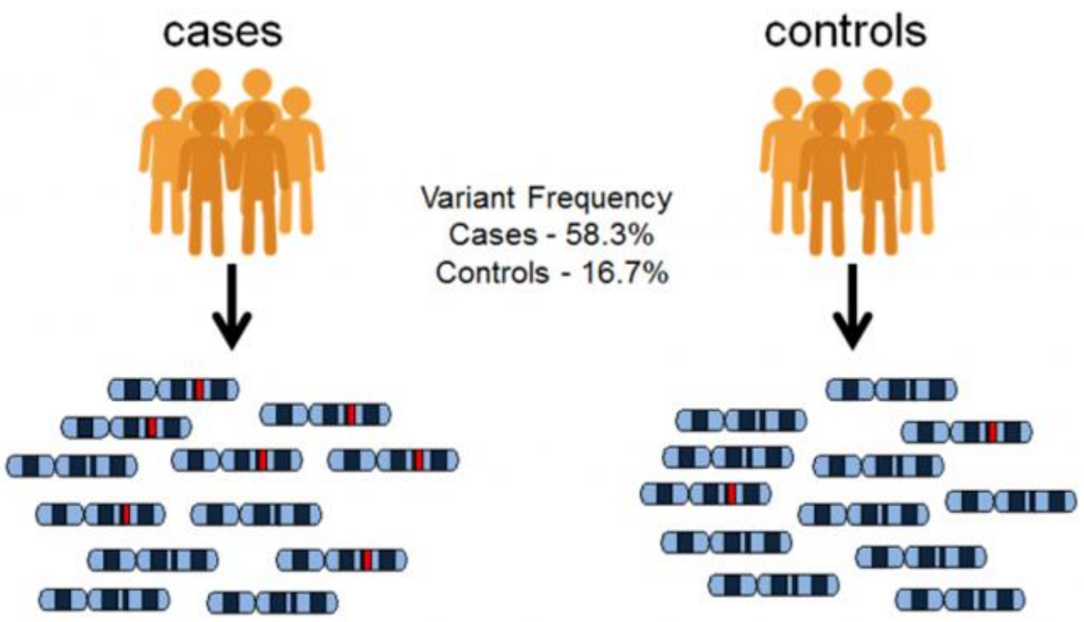
- Information on group of individuals over time
- Exposure variables, biological samples at multiple time points
- Outcome of interest: risk of disease (multiple disease outcomes)
- Size, sampling designs

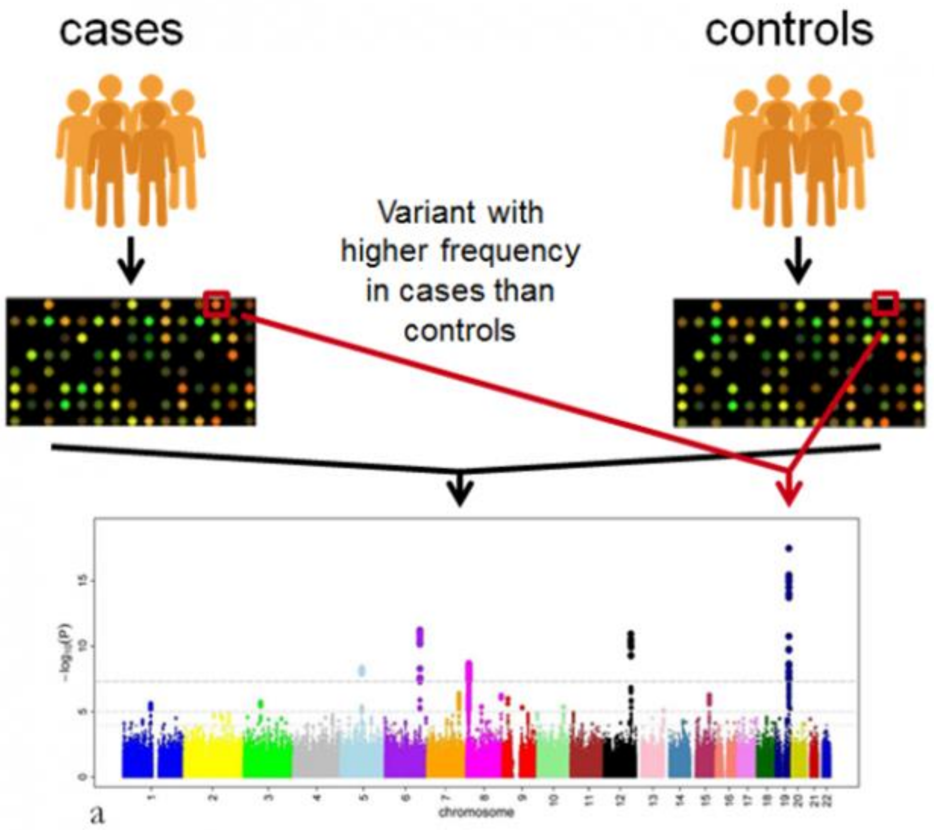
CASE-CONTROL STUDIES



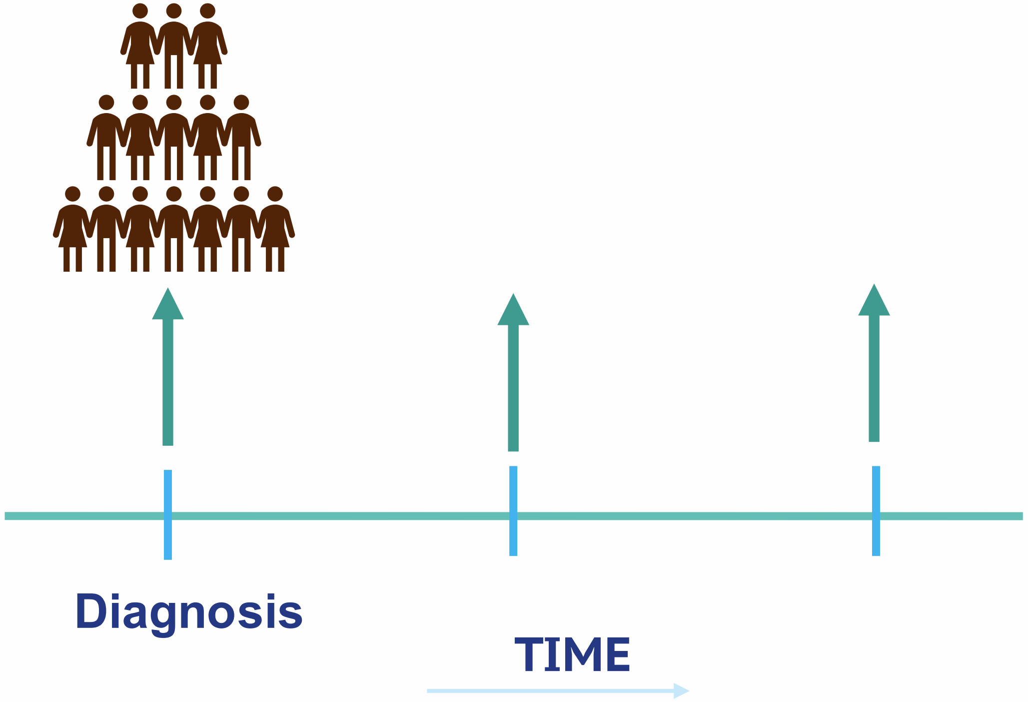
- Sampling of cases and controls from a population source
- Proper case-control selection (population vs. hospital-based controls)
- Selection bias
- Research question to define eligibility of case/controls
- One disease outcome, if multiple disease controls can be shared

CASE-CONTROL STUDIES: GWAS GENETIC SUSCEPTIBILITY





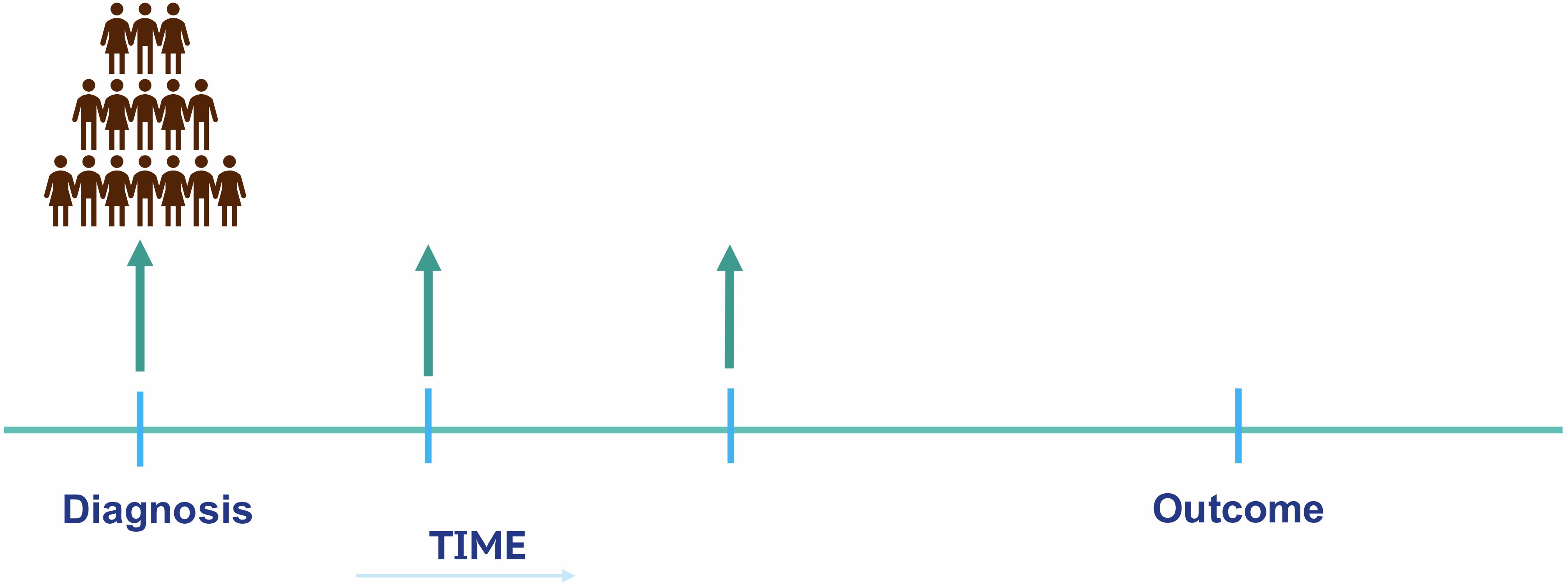
CASE ONLY STUDIES (CASE SERIES)



Information at various timepoints in a short period:

- Exposures variables
- Final and intermediate endpoints
- Disease at a specific time point

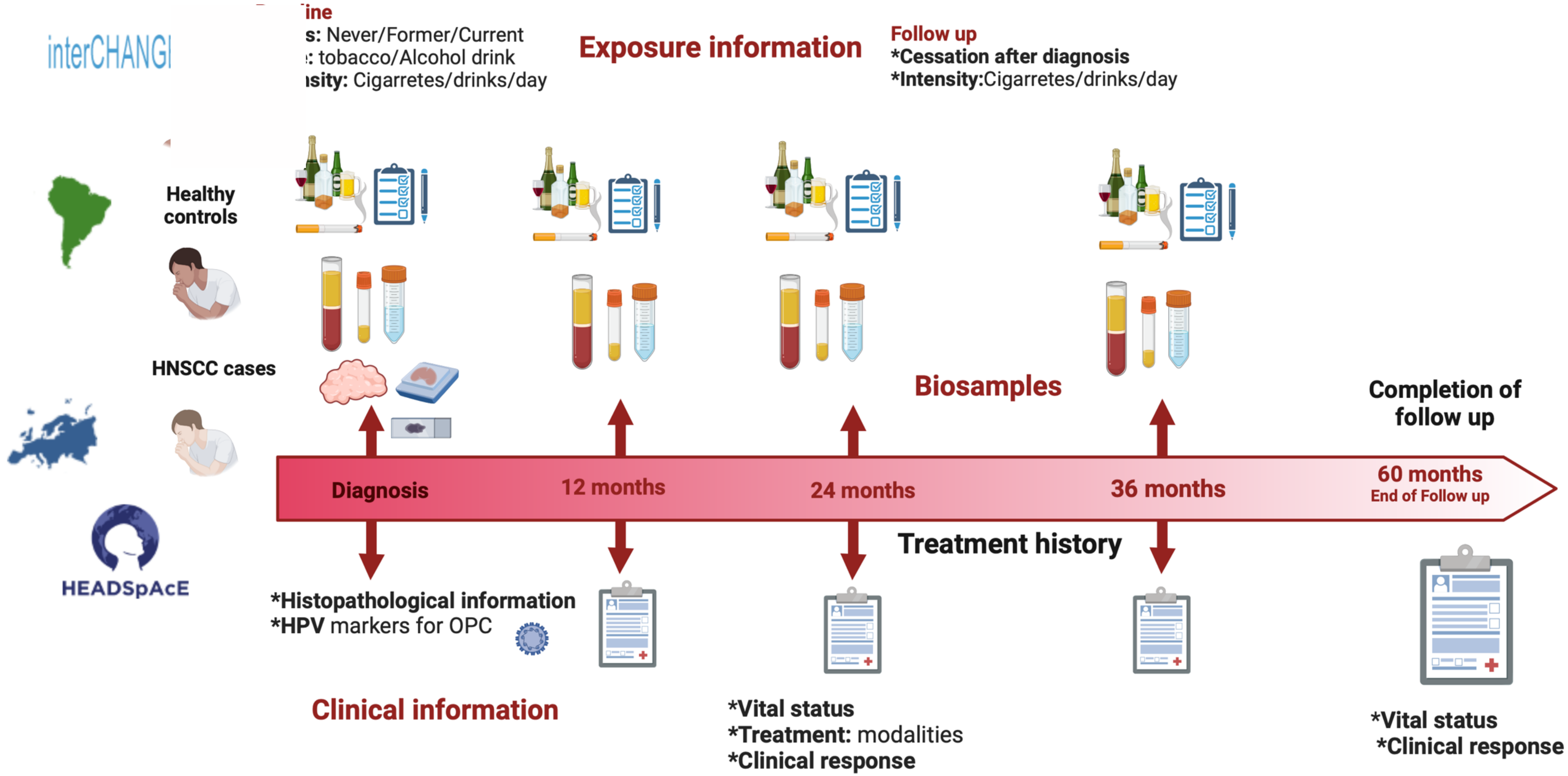
CASE ONLY STUDIES (CASE SERIES)



Information at various timepoints in a short period:

- **Exposures variables: classification of disease based on molecular features and some exposures (*)**
- **Final and intermediate endpoints: disease progression, treatment responses, prognosis and outcomes**
- **Disease at a specific time point**

CASE ONLY STUDIES (CASE SERIES)



International Agency for Research on Cancer
Cancer research into action

CONSIDERATIONS FOR STUDY DESING/IMPLEMENTATION

- There is **no perfect** design, but a **good design** offers better results/outcomes/benefits
- Spend as much time as needed thinking about **the research question** and what you want to obtain from the study
- Planning is important, but balance it with **adaptability**
- Consider location settings, infrastructure, legal, ethics and administrative regulations
- Engage early with the population/researchers/staff
- **Pilot phase is important!!!**
- Resources and sustainability

60 years | International Agency for Research on Cancer

Cancer research into action

Data • Discovery • Prevention • Knowledge